

Francisco Fernández Barroso

<https://fdezbarroso.github.io/> | f.fdezbarroso@gmail.com | (+34) 618-468-115 | Madrid, Spain

Summary

Videogame Programmer with experience working on gameplay systems in Unreal Engine, large C++ production codebases, and tool architecture, currently working in AAA development at MercurySteam. Recently shipped a vertical slice on Steam for *Lilith: Rise of the Fallen*, built in Unreal Engine 5.

Work Experience

MercurySteam - Tools Programmer

Sep 2025 - Present

- Develop and maintain tools for the proprietary Mercury Engine using C++, Lua and Qt, directly enabling the design team to iterate on gameplay content in a AAA production environment.
- Debug engine-level issues impacting runtime behaviour and content workflows across teams.
- Refactor legacy editor systems to improve stability, performance and iteration speed.

Flaming Dogs Studios - Game Programmer (Academic Studio)

Oct 2024 - Sep 2025

- Contributed to the development and release of a polished vertical slice on Steam for the videogame *Lilith: Rise of the Fallen*, built in Unreal Engine 5 as part of a 20+ person team.
- Designed and implemented core gameplay systems in C++ using Unreal's Gameplay Ability System, employing a data-driven architecture based on Data Assets for all main game features.
- Implemented enemy spawning and foundational AI behaviour, collaborating with designers to iterate on combat mechanics, game feel and systems balance.
- Designed a custom UI navigation manager using Enhanced Input, replacing Unreal's default focus system to ensure robust gamepad support across menus.
- Fixed numerous memory leaks and crashes by refactoring the codebase to use Weak Pointers.

Pix4D - Software Engineer

Apr 2022 - Aug 2025

- Developed and maintained complex C++ systems and features within a large production codebase, focusing on performance, correctness, and long-term maintainability.
 - Built automated QA, CI and performance tracking systems in C++ and Python.
 - Improved a core detection algorithm using custom point-cloud filtering techniques, achieving upwards of a 15% F1-score improvement.
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Projects

Project Helltech | Unreal Engine 5, C++, GAS | Movement-focused first person shooter prototype.

CHIP-8 emulator | C++, Qt, SDL2 | Cycle-accurate and feature-complete interpreter for the system.

Gōu | Unity, C# | Atmospheric horror game, developed and designed enemy AI and core interaction systems.

Technologies

Languages and Engines: C++, Lua, Python, C# | Unreal Engine 5 (GAS, Slate), Mercury Engine, Unity.

Tools and Libraries: Perforce, Git, CMake, Jenkins | SDL2, FMOD, QML/Qt, Linux.

Education

Videogame Programming | Master's Degree

Oct 2024 – Sep 2025

Focus: Gameplay and Systems Programming, Unreal Engine, AI, UI | Grade:10/10

Universidad Complutense de Madrid

Computer Engineering | Bachelor's Degree

Sep 2016 – Jun 2022

Universidad Politécnica de Madrid